Answer Partitions~~and~~ , Oracle ~~Access Determine~~ Response Spectra, and Exact Quantum Query ~~Complexity~~Advantage

Karl Svozil *
Institute for Theoretical Physics, TU Wien, Wiedner Hauptstraße 8–10/136, 1040 Vienna, Austria
(Dated: November 4, 2026)

An ~~answer partition specifies which oracle instances share an output, while query complexity also depends on how those instances are accessed. We formulate exact one-query answer-partition resolution as block discrimination of a unitary oracle family. Applied to Deutsch's problem with a controlled response~~ oracle problem has two parts: an answer partition, which says which hidden instances must receive the same answer, and an access model, which says what a query returns. We show that the access model can change the exact quantum query complexity in ways no classical query can detect. One query solves a task exactly if and only if some probe makes the output states of different answer cells orthogonal; we state this criterion together with a certificate that proves when it fails. For an oracle that applies a fixed unitary V, ~~the criterion shows that one query suffices~~ its response, to the target whenever the queried bit is 1, one query solves a dictator (a function equal to a single input bit) exactly if and only if 0 lies in the convex hull of the spectrum of V, and a two-bit parity, including Deutsch's problem, if and only if -1 is an eigenvalue of V; ~~cyclic addition in odd response dimension therefore raises the exact quantum cost to two queries, matching the classical cost while preserving the samecomputational-basis point values. Holding the standard Boolean point oracle fixed, we further show that balanced answer partitions with equal-sized cells can have different exact classical and quantum query complexities . A complete exact classification is given for all~~ no other total Boolean function is one-query exact. Hence a qutrit transposition and a cyclic shift, which return the same values on a target prepared in 0, need one and two exact queries; with bounded error, one query to the shift still succeeds with the optimal probability 7/8. An inaccessible copy of k address bits raises the cost of identifying an n-bit Bernstein–Vazirani string from one query to $\min\{k+1, n\}$, although every classical value stays the same. For the standard oracle we reorganize the known exact complexities of the 35 balanced three-bit partitions ~~, while the unresolved~~ and classify all 6435 balanced four-bit ~~cases are reported explicitly as numerical candidates. These results separate the combinatorial specification of an answer from the oracle-dependent distinguishability that determines query complexity.~~ partitions exactly; rational dual certificates prove the 6040 four-bit lower bounds that were known only from numerical semidefinite programming.

~~An oracleproblem pairs an answer partition with an access model for the hidden instance. The partition groups~~

I. INTRODUCTION

In Deutsch's problem a hidden function maps $x \in \{0,1\}$ to a bit b_x, and one must decide whether it is constant ($b_0 = b_1$) or balanced ($b_0 \neq b_1$). To keep the query unitary, the oracle does not overwrite a register with b_x but adds b_x to a target bit y, $|x, y\rangle \mapsto |x, y \oplus b_x\rangle$; started at the blank value $y = 0$, the target then shows b_x. With this XOR oracle one quantum query suffices, against two classical ones [1–3].

Now take a three-level target $y \in \{0,1,2\}$ and require, for a fair comparison, the same classical readout as for the XOR oracle: started at the blank value $y = 0$, the target shows b_x. This requirement fixes only what happens to $y = 0$; what the oracle does with $y = 1$ and $y = 2$ is left open. When $b_x = 1$, the cyclic shift $y \mapsto y + 1 \bmod 3$ moves $0 \to 1 \to 2 \to 0$, whereas the transposition exchanges 0 and 1 and leaves 2 alone. Classically the two are equally useful: whatever basis state the target starts in, one query reveals at most the bit b_x, so both need two classical queries. Yet with the shift an exact answer needs two quantum queries, with the transposition only one (Sec. III). A quantum algorithm can also prepare the target in an eigenstate of the oracle's response, which the oracle only multiplies by the corresponding eigenvalue. Deutsch's one-query solution needs the eigenvalue -1. The bit flip of the XOR oracle and the transposition both have it, with eigenstate $|0\rangle - |1\rangle$; the eigenvalues of the shift are the cube roots of unity, and none of them is -1. A classical query feeds in a single basis state and never sees these eigenvalues.

In general, a query problem has two parts: an answer partition, which groups the hidden instances into cells whose members share the ~~same required answer. The access modeldetermines what information each query makes available, classically or quantum mechanically, and how that information can be processed. Quantum query speeduptherefore belongs to the task-oracle pair~~correct answer, and an access model, which says what each query returns. Query complexity, and with it any quantum speedup, belongs to this pair, not to the task

* karl.svozil@tuwien.ac.at

alone. That it depends on the access model is known: simulating an in-place oracle for a permutation of N points can require $\Theta(\sqrt{N})$ uses of the standard oracle for the same function [4]; the comparison of quantum and classical oracle power depends on how the oracle is represented [5]; a richer classical oracle solves Deutsch's problem with one query [6]; and a faulty oracle removes Grover's speedup [7]. ~~We make this viewpoint precise and then test it on standard point oracles, response-unitary variants, and small exhaustive families of balanced tasks.~~

~~This viewpoint also sharpens the usual intuition about quantum parallelism. A quantum query prepares one coherent state, and~~ Earlier work, including the author's, posed quantum decision problems for given quantum states: the possible states are partitioned into equivalence classes, and a measurement must find the class of the state at hand; identifying the state completely is the special case of one-element classes [8–13]. Query problems are partitioned in the same way, but no state is given. The hidden instance is a transformation that the algorithm can only call, an oracle or black box. "Oracle" is the computer-science term, going back to Turing's oracle machines [14]: something one puts questions to and that answers. "Black box" stresses that one can only call it, not look inside. We use "oracle" throughout. The algorithm must first produce states by querying it. It succeeds exactly when the ~~algorithm succeeds when interference makes the states associated with different answer cells distinguishable. The useful resource is consequently the structure of the oracle response—its phases, spectrum, accessible registers, and possible environment—rather than simply the formal number of branches in a superposition. Rewriting an oracle by fixed basis changes preserves this discrimination, whereas changing its responserepresentation, aggregating information, or discarding a witness can alter it [5]. The examples that follow develop these ideas through exact one-query discrimination~~states produced for instances with different answers are orthogonal, so that a measurement can tell them apart. This paper asks which properties of the oracle, beyond the classical values it returns, decide how many queries this takes.

The procedure has four steps, used throughout the paper and listed in Sec. VI. First, the task fixes the answer partition: instances with the same correct answer share a cell, and only the cell must be learned. Second, the access model fixes what one oracle call does. Third, an instance-independent probe, typically a superposition fanned out from a single basis state, is sent through the oracle, which turns it into an output state that depends on the hidden instance. Fourth, one call solves the task exactly if and only if some probe makes the output states of different cells orthogonal; overlaps within a cell do no harm. In the example above, this orthogonality needs a relative factor -1 between interfering branches, which the bit flip and the transposition supply and the cyclic shift does not.

The main new result is a spectral criterion (Theorem 1 and Proposition 4). It concerns functions $f(b)$ of a hidden truth table $b = (b_1, \ldots, b_N)$, in particular dictators and parities. A dictator is a function whose value is just one of its input bits, $f(b) = b_i$ (or its negation): one bit dictates the answer and the others do not matter, so classically one query, to address i, suffices. For an oracle that applies a fixed unitary V, its response, ~~parity,~~ to the target whenever the queried bit is 1, one query solves a dictator exactly if and only if 0 lies in the convex hull of the spectrum of V, and a two-bit parity $b_i \oplus b_j$, including Deutsch's problem, if and only if -1 is an eigenvalue of V; no other total Boolean function is one-query exact. For the XOR oracle this reduces to the characterization by Chen, Ye, and Li [15] . It follows that two reversible oracles that agree on the blank value $y = 0$ can have different exact costs, that a phase invisible to basis queries can restore the advantage, and that a cyclic shift on an odd number of levels keeps a bounded-error advantage although it loses the exact one (Sec. III). The second new result is an exact classification of all balanced four-bit tasks for the standard oracle: rational dual certificates prove the two-query lower bounds that were previously known only numerically [16] (Sec. IV). Section II sets up the framework, a block form of standard discrimination conditions and its dual. Section V treats an oracle that leaves an inaccessible copy of the address; this changes Bernstein–Vazirani ~~, and a finite census of balanced partitions~~costs although every classical value stays the same. Table I separates prior results from new ones.

These results also correct a common picture of quantum parallelism: what matters for the separations below is how the oracle responds (its phases, spectrum, accessible registers, and environment), not how many branches a superposition has.

II. ANSWER PARTITIONS AND EXACT ONE-QUERY RESOLUTION

Let Ω be a finite ~~instance set~~ set of instances and let $g : \Omega \to J$ ~~specify the required~~ give the correct answer. The task ~~induces~~ defines the *answer partition* ~~(also called here the kinematic partition)~~

$$\Pi_g = \{C_j : j \in g(\Omega)\}, \qquad C_j = g^{-1}(j). \tag{1}$$

~~This partition is indispensable bookkeeping, but it is not yet an access model or a complexity statement~~The partition says what must be found, not how. In the query model the hidden instance labels ~~a black-box transformation~~an oracle; the algorithm is not ~~handed~~ given a state $|a\rangle$ ~~on which the answer partition is a pre-existing observable. The operational~~ that it could simply measure. The question is whether the ~~specified~~ given oracle and a fixed protocol can ~~transfer~~ bring the cell label ~~to an accessible output~~into a register that can be measured.

TABLE I. Relation to prior work. "Known" lists results available before this paper; "This work" lists what is added here. Q_E is the exact quantum query complexity, Par_N the parity of N bits, V the response unitary, $\mathrm{conv}(\mathrm{spec}\,V)$ the convex hull of its spectrum, d the dimension of the cyclic response $X_d|y\rangle = |y+1 \bmod d\rangle$, and k the number of the n address bits recorded by the witness.

Topic	Known	This work
Exact one-query criterion	Perfect discrimination of unitaries [17, 18]	Block form for answer partitions and its dual certificate (Props. 1 and 2); standard convex duality, stated for use below
One-query exact class	XOR oracle: dictators and two-bit parities [15]; partial functions [19]	Every response unitary V: dictators iff $0 \in \mathrm{conv}(\mathrm{spec}\,V)$, two-bit parities iff $-1 \in \mathrm{spec}\,V$, no other total Boolean functions (Thm. 1, Prop. 4)
Dependence on the oracle model	Oracle representation [5]; in-place versus standard oracles [4]; de-quantised Deutsch oracle [6]; faulty oracles [7]	A spectrum invisible to classical queries decides the exact cost: two reversible oracles that agree on $y=0$, shift with an added phase (Cor. 1); sign oracles cannot separate complementary truth tables (Prop. 3)
Bounded error, odd cyclic response	—	Optimal unambiguous one-query solution, worst-case success $(3+\cos\frac{\pi}{d})/4$ (Prop. 5)
Parity	$Q_E(\mathrm{Par}_N) = \lceil N/2 \rceil$ [20, 21]	Degree bound for every response; open qutrit question (Sec. III)
Three-bit functions	Explicit optimal exact algorithms for all functions [16]	No new values; reorganization as balanced partitions (Table II)
Four-bit functions	Numerical SDP values for all functions [16], used in structural work [22]	Exact rational dual certificates for all 6435 balanced partitions (Table III)
Decohered access	Which-way information [23]; coherence in Bernstein–Vazirani [24]	Exact costs with an inaccessible witness: surviving partition in one query versus $n-k$; identification in exactly $\min\{k+1, n\}$ queries (Prop. 6)

~~More explicitly, let~~ Let $\rho_a^{(T)}$ be the ~~accessible state produced from instance a~~ state of the measurable registers after T queries on instance a. Exact resolution means that ~~there is~~ one instance-independent measurement $\{P_j\}$ ~~satisfying~~

$$P_j P_\ell = \delta_{j\ell} P_j,\ \sum_j P_j = I,\ \mathrm{Tr}(P_{g(a)}\rho_a^{(T)}) = 1.$$

satisfies

$$P_j P_\ell = \delta_{j\ell} P_j, \quad \sum_j P_j = I, \quad \mathrm{Tr}(P_{g(a)}\rho_a^{(T)}) = 1. \tag{2}$$

~~Consequently, supports belonging to different cells are orthogonal ; no orthogonality is required~~ States from different cells then have orthogonal supports; states within a cell ~~. The~~ need not be orthogonal. The cells C_j ~~themselves remain~~ are sets of classical instances~~. They are not subspaces, and the answer sectors~~, not subspaces; the subspaces $P_j\mathcal{H}_{\mathrm{out}}$ ~~arise~~ of the output space $\mathcal{H}_{\mathrm{out}}$ exist only after an oracle and ~~protocol have encoded~~ a protocol have produced them. The partition ~~therefore specifies a required zero pattern in the output distinguishability matrix rather than~~ thus prescribes which output overlaps must vanish; it is not a ready-made ~~physical~~ observable.

~~We next characterize exact one-query resolution for~~ For a partition Π, a quantum oracle family $\mathcal{U}$, and the corresponding classical oracle $\mathcal{U}_{\mathrm{cl}}$, $Q_\epsilon(\Pi;\mathcal{U})$ and $R_\epsilon(\Pi;\mathcal{U}_{\mathrm{cl}})$ denote the smallest worst-case numbers of quantum and classical calls that achieve worst-case error at most ϵ, with matching access on both sides; $Q_E \equiv Q_0$ is the exact quantum complexity, and D is the worst-case number of queries of an exact deterministic classical algorithm (the depth of an optimal decision tree). With worst-case counting, $D = R_0$: fixing the random choices of a zero-error randomized protocol that never uses more than T queries leaves a deterministic protocol that is always correct with at most T queries. (Counted in expectation, as is common, zero-error cost can be smaller than D.) A deterministic classical protocol

groups the instances by the transcript of answers they produce; it succeeds exactly only if instances with the same transcript always lie in the same cell. A quantum protocol keeps no such intermediate record; it changes the overlaps among the instance states and measures only after the cells have separated. A speedup is a gap between the quantum and the classical cost for the same task and the same access.

All Hilbert spaces are finite dimensional. For a family of unitary oracles $\{U_a : a \in \Omega\}$. An arbitrary , a general one-query protocol may prepare prepares a pure state $|\Psi\rangle$ of the query system Q and a reference system R, apply applies $U_a \otimes I_R$, and then use an instance-independent applies a fixed channel and measurement. Write

$$|\Psi_a\rangle = (U_a \otimes I_R)\,|\Psi\rangle\,, \qquad \sigma = \mathrm{Tr}_R\,|\Psi\rangle\langle\Psi|\,. \tag{3}$$

The output Gram matrix before the final processing Gram matrix of the output states is

$$G_{aa'} = \langle\Psi_a|\Psi_{a'}\rangle = \mathrm{Tr}\big(\sigma U_a^\dagger U_{a'}\big)\,. \tag{4}$$

. *Proposition 1 (block discrimination).* The partition Π_g is exactly resolvable with at most one use of

Proposition 1 (Block discrimination)**.** *One query to* $\{U_a\}$, *allowing an arbitrary reference , with any reference system allowed, resolves* Π_g *exactly if and only if there is a some density operator* σ *such that satisfies*

$$\mathrm{Tr}\big(\sigma U_a^\dagger U_{a'}\big) = 0 \quad \textit{whenever} \quad g(a) \neq g(a')\,. \tag{5}$$

Indeed, exact

Proof. Exact discrimination requires the spans of the output states associated with output states of different cells to be orthogonal , which isequivalent to the indicated pairwise inner products vanishing.span orthogonal subspaces, that is, Eq. (5) for every pair from different cells; no fixed channel can make nonorthogonal states perfectly distinguishable. Conversely, orthogonal cell spans can be measured by their support projectors . Purification shows that subspaces are distinguished by the projectors onto them. By purification, optimizing over σ already includes covers mixed probes and arbitrary ancillary systems. ancillas. □

Equation (5) , with $\sigma \succeq 0$ and $\mathrm{Tr}\,\sigma = 1$, is the direct block-discrimination form of standard perfect unitary-discrimination conditions is the block form of the standard conditions for perfect discrimination of unitaries [17, 18]; it is not a new general channel-discrimination theorem . Its role here is to connect theorem about channel discrimination. Here it links an answer partition to a specified given oracle family. The Gram matrix need only be block diagonal according with respect to Π_g; states within one cell may overlap or coincide. Complete identification is the singleton-cell special case Identifying the instance completely is the special case of one-element cells, where G must be diagonal.

For Since zero queries cannot resolve a nontrivial partition, Eq. (5) characterizes $Q_E(\Pi_g;\mathcal{U}) = 1$. With bounded error, orthogonality is replaced by a common POVM with measurement that reaches the required success probability. Multiquery Lower bounds for several queries come from adversary methods [25], whose general form characterizes state conversion and function evaluationare characterized by adversary methods [26, 27] [26, 27].

Write $Q_\epsilon(\Pi;\mathcal{U})$ and $R_\epsilon(\Pi;\mathcal{U}_{\mathrm{cl}})$ for the optimal quantum and classical worst-case numbers of calls needed with worst-case error at most ϵ, under explicitly matched access conventions, and write $Q_E \equiv Q_0$ for exact quantum complexity.Equation (5) characterizes exact resolvability with at most one oracle use, namely $Q_E(\Pi_g;\mathcal{U}) \leq 1$. For every nontrivial partition, zero queries cannot resolve the task, so this is equivalent to $Q_E(\Pi_g;\mathcal{U}) = 1$; the trivial one-cell partition instead has $Q_E(\Pi_g;\mathcal{U}) = 0$. The criterion does not by itself establish an advantage. Speedup additionally requires a gap between Q_ϵ and R_ϵ for the same task and oracle interface. Classically, a deterministic query protocol partitions Ω by its possible transcripts and succeeds exactly only when that transcript partition refines Π_g. A quantum protocol need not create such an intermediate classical transcript; it changes the overlaps among the hypothesis states and measures only after the required cellshave separated. The relevant resource is this rate of separation, not the number of formal components in When Eq. (5) has no solution, a single matrix proves it.

Proposition 2 (Dual certificate)**.** *Let* $X = \{(a,a') : g(a) \neq g(a')\}$. *Exactly one of the following holds: (i) a density operator* σ *satisfies Eq. (5); (ii) there is a complex matrix* Γ, *supported on* X *and satisfying* $\Gamma_{a'a} = \Gamma^*_{aa'}$, *such that*

$$Z(\Gamma) = \sum_{(a,a')\in X} \Gamma_{aa'}\, U_a^\dagger U_{a'} \succ 0. \tag{6}$$

Proof. $Z(\Gamma)$ is Hermitian because $(U_a^\dagger U_{a'})^\dagger = U_{a'}^\dagger U_a$. If (i) and (ii) both held, then $\mathrm{Tr}[\sigma Z(\Gamma)] = 0$, contradicting $Z(\Gamma) \succ 0$. If (ii) fails, the real subspace $\{Z(\Gamma)\}$ of Hermitian operators misses the open convex cone of positive definite operators, so a separating hyperplane gives a nonzero σ with $\mathrm{Tr}(\sigma P) \geq 0$ for all $P \succ 0$, that is $\sigma \succeq 0$, and $\mathrm{Tr}[\sigma Z(\Gamma)] = 0$ for all admissible Γ. Taking Γ supported on a single pair (a,a') and its transpose, with real and with imaginary entries, gives $\mathrm{Tr}(\sigma U_a^\dagger U_{a'}) = 0$; normalizing σ gives (i). □

Like an adversary matrix, Γ weights only pairs from different cells. After t queries the same weights define the progress measure $\sum_{(a,a')\in X} \Gamma_{aa'} \langle\psi_a^{(t)}|\psi_{a'}^{(t)}\rangle$ of the adversary method [25], and the probe superpositionquery cost measures how fast such weighted cross-cell overlaps

can be driven to zero; this is what it means for a quantum algorithm to "separate" answer cells. When all $U_a^\dagger U_{a'}$ commute, as for the oracles of Secs. III and IV, which are diagonal in the eigenbasis of their response, only the diagonal of σ in a common eigenbasis enters Eq. (5), and the criterion becomes a linear feasibility problem for a probability vector.

The criterion also ~~distinguishes a~~ separates a mere change of representation from a change of access. If

$$V_a = A U_a B \tag{7}$$

for ~~instance-independent~~ fixed unitaries A, B, then $V_a^\dagger V_{a'} = B^\dagger U_a^\dagger U_{a'} B$~~; as the probe varies, the attainable Gram matricesare unchanged~~, so both families reach the same Gram matrices. More generally, families that ~~exactly simulate one another~~ simulate each other exactly with $O(1)$ ~~oracle calls have quantum query complexities equal up to constant factors~~calls have the same quantum query complexity up to a constant factor; an approximate simulation ~~must also charge its conversion~~ adds its error to ϵ. ~~Preservation of a claimed~~ For a quantum–classical gap ~~additionally requires corresponding constant-query classical simulations. Exact one-query behavior can nevertheless change when the response representation is not of the form~~ to carry over, the classical oracles must simulate each other in the same way. Families not related by Eq. (7) can differ in their exact one-query cost, as the next section shows.

~~For example, Hadamard conjugation of the answer qubit turns~~

III. THE RESPONSE SPECTRUM DECIDES EXACT ONE-QUERY POWER

A. Deutsch's problem and the sign oracle

Let $b = (b_0, b_1)$ be the truth table of a Boolean function and use, with address x and target bit y,

$$O_b\,|x, y\rangle = |x, y \oplus b_x\rangle\,. \tag{8}$$

With the address in $|+\rangle$ and the target qubit in $|-\rangle$, phase kickback leaves the address register in the states $|\psi_b\rangle$, labelled by the truth table $b = b_0 b_1$ (the target qubit factors out),

$$|\psi_{00}\rangle = -\,|\psi_{11}\rangle = |+\rangle\,, \quad |\psi_{01}\rangle = -\,|\psi_{10}\rangle = |-\rangle\,. \tag{9}$$

Constant and balanced functions give orthogonal states, while the two functions within each cell give the same state up to sign. The Gram matrix is block diagonal but not diagonal, and the result is the familiar separation of one quantum versus two classical queries [1–3].

Hadamard gates on the target qubit turn Eq. (8) ~~below into the full controlled-phase action~~ into the controlled phase $|x, z\rangle \mapsto (-1)^{z b_x}\,|x, z\rangle$, so the two forms are ~~exactly equivalent . The restricted~~ equivalent in the sense of Eq. (7). The sign oracle $P_b\,|x\rangle = (-1)^{b_x}\,|x\rangle$~~is only a subroutine: $P_{\bar b} = -P_b$, and the complementary truth tables differ~~, without the control qubit z, is weaker. A basis query to it reveals nothing about b_x, and for the complemented truth table $\bar b$ one has $P_{\bar b} = -P_b$, so the two differ only by a global phase. ~~Supplying a coherent control for~~

Proposition 3 (Complement obstruction)**.** *With access to $\{P_b\}$ alone (any ancillas, but no control on P_b~~restores the missing relative phase, but coherent control is additional black-box access rather than an automatic operation on an unknown unitary [28]. ~~), no number of queries resolves a partition that puts some b and its complement $\bar b$ in different cells, even with error $\epsilon < 1/2$.*

Proof. Each use of $P_b \otimes I$ equals $-(P_b \otimes I)$, and every branch of an adaptive protocol picks up the same sign. The final density operators for b and $\bar b$ are therefore identical, and so are the output distributions. □

Deutsch's ~~problem illustrates both the block criterion and this dependence. Let $b = (b_0, b_1)$ be the truth table of a Boolean function and use~~

$$O_b\,|x, y\rangle = |x, y \oplus b_x\rangle\,.$$

~~With the answer qubit in $|-\rangle$, phase kickback gives, up to the common answer factor, the output states on the address register:~~

$$|\psi_{00}\rangle = |+\rangle\,, \quad |\psi_{11}\rangle = -\,|+\rangle\,,$$
$$|\psi_{01}\rangle = |-\rangle\,, \quad |\psi_{10}\rangle = -\,|-\rangle\,.$$

~~Thus the constant and balanced cells occupy orthogonal rays~~constant–balanced partition is closed under complement and is still solved by one query to P_b. Three-bit majority, the parity of an odd number of bits, and every other self-dual function cannot be resolved at all. Of the balanced partitions in Sec. IV, only 3 of 35 for three bits (the two-bit parities) and 35 of 6435 for four bits are closed under complement; the sign oracle resolves none of the others, while the ~~two instances inside either cell are not distinguished. This is precisely a block-diagonal, but not diagonal, Gram matrix and gives the familiar one versus two exact-query separation [1]~~XOR oracle resolves all of them with at most two ($N = 3$) or three ($N = 4$) queries. A coherent control on P_b would restore the missing relative phase, but control is extra access to the oracle, not something one can add to an unknown unitary [28].

~~The phenomenon is controlled by the response spectrum.~~

B. Response unitaries

Let V be any unitary and consider the ~~controlled-response family~~family, indexed by the same four ~~Boolean~~ truth tables, that applies V to the target whenever $b_x = 1$:

$$U_b^{(V)} = \sum_{x=0}^{1} |x\rangle\langle x| \otimes V^{b_x}. \tag{10}$$

. ~~*Theorem 1 (response spectrum).* The Deutsch constant balanced partition is exactly one-query resolvable through~~ *Access conventions.* V is known. Between queries the algorithm may apply any fixed operation, including preparing eigenvectors of V, at no query cost; it may not control, invert, or otherwise modify the oracle unless that is explicitly allowed. The phase of V matters: V^{b_x} acts only on the branch selected by the address, so replacing V by $e^{i\theta}V$ multiplies Eq. (10) by $e^{i\theta b_x}$, a phase that differs between address branches and is therefore observable; it can change the spectral condition below (Corollary 1). This differs from access to an unknown unitary or channel, which is defined only up to a global phase.

Theorem 1 (Response spectrum)**.** *One query to* $\{U_b^{(V)}\}$*, with ~~arbitrary probes and references,~~ any probe and reference, solves Deutsch's problem exactly if and only if* -1 *is an eigenvalue of* V*.*

~~For necessity, write the probe as $|0\rangle|\phi_0\rangle + |1\rangle|\phi_1\rangle$, with $|\phi_x\rangle$ on the response and reference systems.Let $\rho_x = \mathrm{Tr}_R |\phi_x\rangle\langle\phi_x| \succeq 0$ be its unnormalized response operator and put~~

$$p_x = \mathrm{Tr}\,\rho_x, \qquad z_x = \mathrm{Tr}(\rho_x V), \qquad p_0 + p_1 = 1.$$

~~The~~

Proof. If $V|w\rangle = -|w\rangle$, the probe $|+\rangle|w\rangle$ gives ordinary phase kickback and Eq. (9). Conversely, give weight $\Gamma = 1$ to the four cross-cell ~~orthogonality conditions include their complex conjugates and reduce to~~

$$z_0 = z_0^* = -p_1, \qquad z_1 = z_1^* = -p_0.$$

pairs (and their transposes) in Proposition 2. The pairs $(00,01)$, $(00,10)$, $(11,01)$, and $(11,10)$ contribute $I \oplus V$, $V \oplus I$, $V^\dagger \oplus I$, and $I \oplus V^\dagger$ on the address blocks $x = 0, 1$. Together with their adjoints, each address block receives $4I + 2V + 2V^\dagger$, and since V is unitary,

$$Z = 2\,I_2 \otimes (2I + V + V^\dagger) = 2\,I_2 \otimes (I + V)(I + V)^\dagger. \tag{11}$$

~~Since $|z_x| \le p_x$, these equations force $p_0 = p_1 = 1/2$ and $z_0 = z_1 = -1/2$. Equality in the unitary expectation bound is possible only when the support of each nonzero ρ_x lies in the~~ Z is positive definite unless $I + V$ is singular, that is, unless -1 ~~eigenspace of~~ is an eigenvalue of V~~. Conversely, a -1 eigenvector provides ordinary phase kickback and Eq. (9).~~ . □

~~Related qudit~~ In words: at each of the two addresses the probe sees an average of eigenvalues of V, a point of $\mathrm{conv}(\mathrm{spec}\,V)$, and orthogonality across cells forces both points to -1. Since -1 lies on the unit circle, an average of eigenvalues equals -1 only if -1 is itself an eigenvalue. Qudit versions of Deutsch–Jozsa ~~use higher-dimensional~~ enlarge the address and response systems ~~[29]. The theorem above instead holds~~ [2, 3, 29] ; Theorem 1 instead keeps the four Deutsch instances fixed and varies the response unitary. It ~~is also distinct from prior-distribution~~ also differs from "useless-query" bounds, which ~~infer~~ derive quantum lower bounds, under a prior distribution, from classical queries that reveal ~~no information~~ nothing about a property [30].

. ~~*Corollary 1 (cyclic response).* For the elementary cyclic response $X_d|y\rangle = |y+1 \bmod d\rangle$, $d \ge 2$,~~

Corollary 1 (Same classical values, different cost)**.** *On a qutrit, consider the transposition* $|0\rangle \leftrightarrow |1\rangle$ *that fixes* $|2\rangle$*, the cyclic shift* $X_3|y\rangle = |y+1 \bmod 3\rangle$*, and* $X_3' = e^{i\pi/3}X_3$*. All three map the blank target* $|0\rangle$ *to* $|1\rangle$ *up to a phase* ($X_3'|0\rangle = e^{i\pi/3}|1\rangle$)*, so measuring a blank target returns* b_x *in each case;* X_3 *and* X_3' *moreover agree on every computational-basis ~~queries return the same Boolean point value for every d (by mapping $y \mapsto y + b_x \pmod d$)~~ input up to a phase. Their spectra are* $\{1, 1, -1\}$*,* $\{1, e^{2\pi i/3}, e^{4\pi i/3}\}$*, and* $\{e^{i\pi/3}, -1, e^{-i\pi/3}\}$*, so the exact quantum cost of Deutsch's problem is* $Q_E = 1$*,* 2*, and* 1*, respectively, while* $D = 2$ *in all three cases. More generally, for* $X_d|y\rangle = |y+1 \bmod d\rangle$ *with ~~$b_x \in \{0,1\}$), but -1 occurs in the spectrum of X_d exactly when d is even. Therefore~~* $d \ge 2$*,*

$$Q_E^{(d)} = \begin{cases} 1, & d \text{ even}, \\ 2, & d \text{ odd}, \end{cases} \qquad D^{(d)} = 2. \tag{12}$$

~~*In particular, reversible qutrit addition removes*~~

Specifying an oracle by what it does to a blank target, $|x\rangle|0\rangle \mapsto |x\rangle|b_x\rangle$, fixes only one column of each V^{b_x}. How the rest is filled in, by a transposition or a cyclic shift, decides whether Deutsch's exact ~~quantum advantage without changing its classical point values . This is a constant-factor effect~~advantage exists, and so does a phase that no basis query can see: the factor $e^{i\pi b_x/3}$ in $X_3'^{b_x}$ differs between address branches; it is a phase oracle composed with the shift. Neither the classical values nor even the full action on basis states up to phases determines Q_E. The effect is a constant factor, not an asymptotic separation~~, but it demonstrates that reversibility and blank-register behavior do not determine coherent query power~~.

~~Parity provides a broader illustration of~~

C. Any number of bits

The same argument works for any number of bits. Let $f:\{0,1\}^N\to\{0,1\}$ be nonconstant, with oracle $U_b^{(V)}=\sum_{i=1}^N|i\rangle\langle i|\otimes V^{b_i}$. Below, e_i is the truth table with a single 1 at address i, and 0^N the ~~same principle under standard pointwise access.~~ all-zero truth table.

Proposition 4 (One-query class)**.** *One query to $\{U_b^{(V)}\}$ resolves a nonconstant f exactly if and only if, up to negation, either $f=b_i$ and $0\in\operatorname{conv}(\operatorname{spec}V)$, or $f=b_i\oplus b_j$ and $-1\in\operatorname{spec}V$. The dictator condition $0\in\operatorname{conv}(\operatorname{spec}V)$ holds in particular if $V|y\rangle\perp|y\rangle$ for some basis state $|y\rangle$.*

Proof. Let ~~$b\in\{0,1\}^N$ and $O_b|i,y\rangle=|i,y\oplus b_i\rangle$. Every~~ $\rho_i\succeq 0$ be the diagonal block of σ at address i, an operator on the target, and $p_i=\operatorname{Tr}\rho_i$, so $\sum_i p_i=1$. Since $U_b^\dagger U_{b'}=\sum_i|i\rangle\langle i|\otimes V^{b'_i-b_i}$, only the ρ_i enter Eq. (5). If f depends on b_i, some b has $f(b)\neq f(b\oplus e_i)$, and Eq. (5) reads $1-p_i+\operatorname{Tr}(\rho_iV^{\pm1})=0$. Since $|\operatorname{Tr}(\rho_iV^{\pm1})|\le p_i$, this forces $p_i\ge 1/2$, so f depends on at most two bits. If it depends on two, i and j, then $p_i=p_j=1/2$ and $\operatorname{Tr}(\rho_iV^{\pm1})=-p_i$. An average of unit-modulus eigenvalues equals -1 only if all its weight lies on the eigenvalue -1, so ρ_i and ρ_j are supported on the -1 eigenspace of V. The overlap of the outputs for b and $b\oplus e_i\oplus e_j$ is then $-1\neq 0$, so these instances share a cell, $f(b)=f(b\oplus e_i\oplus e_j)$, and f is $b_i\oplus b_j$ or its negation. A -1 eigenvector gives the Deutsch construction on addresses i,j. For a dictator b_i, the cross-cell pairs $(0^N,e_i)$ and $(0^N,e_i\oplus e_j)$, $j\neq i$, give $\operatorname{Tr}(\rho_iV)+\sum_{k\neq i}p_k=0$ and, after subtracting, $\operatorname{Tr}(\rho_jV)=p_j$ for every $j\neq i$. The density operator $\tau=\sum_k\rho_k$ therefore has $\operatorname{Tr}(\tau V)=0$, that is, $0\in\operatorname{conv}(\operatorname{spec}V)$. Conversely, the probe $|i\rangle|w\rangle$ with $\langle w|V|w\rangle=0$ makes every cross-cell overlap vanish. With an additional null address, on which the oracle acts as the identity (as in the phase form used in Sec. IV), the same argument holds with $\operatorname{spec}V$ replaced by $\operatorname{spec}V\cup\{1\}$ in the dictator condition. □

For a dictator b_i it suffices to query address i alone; one query then succeeds exactly when some target state $|w\rangle$ is turned into an orthogonal one, $\langle w|V|w\rangle=0$, that is, when an average of eigenvalues of V vanishes. For $V=-I$, the family is the sign oracle of Proposition 3: one query resolves two-bit parities, but no number of queries resolves a dictator.

For the XOR oracle, $V=X$, Proposition 4 reproduces the characterization of exact one-query algorithms by Chen, Ye, and Li [15]. Proposition 4 concerns total functions. With a promise, more tasks can be one-query exact, the Deutsch–Jozsa problem [2] being the standard example; partial functions are treated in Ref. [19]. For an odd cyclic response X_d only the dictators remain, and a single classical query already resolves them.

D. Bounded error for odd cyclic responses

Corollary 1 concerns exact solutions. With bounded error, the odd cyclic response keeps a quantum advantage. As a referee pointed out, a single eigenvector of X_3 with eigenvalue $e^{2\pi i/3}$, the address in $|+\rangle$, and a measurement of the address in $\{|+\rangle,|-\rangle\}$ already give success 1 on constant and 3/4 on balanced instances, whereas one classical query cannot exceed worst-case success 1/2. Spreading the probe over the two eigenvalues adjacent to -1 raises the worst case further and makes every conclusive answer correct.

Proposition 5 (Optimal one-query solution)**.** *For odd $d\ge 3$, one query to $\{U_b^{(X_d)}\}$ solves Deutsch's problem with zero error and a conclusive answer with probability $(1+\cos\frac{\pi}{d})/2$ on every instance. Guessing on the inconclusive outcome gives worst-case success $(3+\cos\frac{\pi}{d})/4$, that is 3/4 conclusive and 7/8 overall for $d=3$, and no one-query protocol does better. One classical query never gives a conclusive zero-error answer and has worst-case success 1/2.*

Proof. Let $|v_\pm\rangle$ be eigenvectors of X_d with the eigenvalues $\lambda_\pm=-e^{\pm i\pi/d}$ adjacent to -1, and use the probe $\frac12\sum_{x=0}^1\sum_\pm|x\rangle|v_\pm\rangle$. Writing $\gamma=\cos\frac{\pi}{d}$ and using $(\lambda_+^k+\lambda_-^k)/2=-\gamma$ for $k=\pm1$, the output Gram matrix is $G_{bb'}=\frac12\sum_x c_x$ with $c_x=1$ if $b_x=b'_x$ and $c_x=-\gamma$ otherwise. Hence $|\psi_{00}\rangle+|\psi_{11}\rangle=|\psi_{01}\rangle+|\psi_{10}\rangle$, with squared norm $2(1-\gamma)$, while $|\psi_{00}\rangle-|\psi_{11}\rangle$ and $|\psi_{01}\rangle-|\psi_{10}\rangle$ are orthogonal to each other and to that common vector, each with squared norm $2(1+\gamma)$. Projecting onto these three directions returns "constant," "balanced," or "inconclusive"; each instance gives the correct conclusive outcome with probability $(1+\gamma)/2$ and never the wrong one.

For optimality, write any pure probe, including a reference~~system, can be written in the answer qubit's X basis as~~

$$|\Psi\rangle=\sum_{i=1}^N|i\rangle\left(|+\rangle|\alpha_i\rangle+|-\rangle|\beta_i\rangle\right),\qquad p_i=\|\beta_i\|^2.$$

~~The overlap between the outputs for 0^N and the odd string e_i is~~

$$\langle\Psi_{0^N}|\Psi_{e_i}\rangle=1-2p_i.$$

~~Exact even-odd separation would require $p_i=1/2$ for every i, while normalization gives $\sum_i p_i\le 1$. Hence one query is possible only for $N\le 2$,~~ as $|0\rangle|A\rangle+|1\rangle|B\rangle$ with ~~Deutsch as the limiting case. More generally, $\|A\|^2+\|B\|^2=1$, and put~~ $W=V\otimes I_R$ with $V=X_d$. Let ρ_C and ρ_B be the ~~acceptance probability of a~~ outputs averaged uniformly over the constant and over the balanced instances. Their diagonal address blocks coincide, and their

off-diagonal blocks differ by $\frac{1}{2}|(I-W)A\rangle\langle(I-W)B|$, so $\|\rho_C-\rho_B\|_1=\|(I-W)A\|\,\|(I-W)B\|\le\frac{1}{2}\|I-V\|^2$, where $\|\cdot\|_1$ is the trace norm and $\|\cdot\|$ the operator norm (for vectors, the Euclidean norm). Worst-case success cannot exceed success under this prior, which by the Helstrom bound is at most $\frac{1}{2}+\frac{1}{8}\|I-V\|^2$. For odd d, $\|I-X_d\|^2=2+2\cos\frac{\pi}{d}$, which gives $(3+\cos\frac{\pi}{d})/4$. Classically, one query reveals one bit b_x, and both cells remain consistent with every transcript. □

The bound also makes the construction optimal among unambiguous protocols. If every instance gave a conclusive answer with probability at least p, guessing on the inconclusive outcome would give worst-case success at least $(1+p)/2$; hence $p\le(1+\cos\frac{\pi}{d})/2$, which the construction attains.

The bound $\frac{1}{2}+\frac{1}{8}\|I-V\|^2$ holds for every response V and reaches 1 only if $\|I-V\|=2$, that is, only if -1 is an eigenvalue; this is a second proof of the necessity part of Theorem 1. The construction parallels Deutsch's original algorithm, which also answered conclusively only part of the time [1]; here the inconclusive probability $(1-\cos\frac{\pi}{d})/2$ is set by how close the spectrum comes to -1, and it vanishes as $d\to\infty$.

E. Parity

Proposition 4 shows that one query resolves the parity $\mathrm{Par}_N(b)=b_1\oplus\cdots\oplus b_N$ only for $N\le 2$. Proposition 2 gives an explicit witness for the XOR oracle: with $\Gamma=1$ on the pairs $(0^N,e_i)$,

$$Z=\sum_{i=1}^{N}\left(O_{e_i}+O_{e_i}^{\dagger}\right)=2N\,I-4\Pi_{-},\tag{13}$$

where $\Pi_-=\sum_i|i\rangle\langle i|\otimes|-\rangle\langle-|$; Z is positive definite exactly when $N>2$. For more queries, note that $U_b^{(V)}=\sum_i|i\rangle\langle i|\otimes[I+b_i(V-I)]$ is affine in b for every response V. The amplitudes after T ~~query algorithms are a polynomial~~ queries are therefore polynomials of degree at most ~~$2T$~~ T in the input bits, ~~whereas~~ the acceptance probability has degree at most $2T$, and parity has exact degree N. Thus $T\ge N/2$ ~~, and~~ for every response, and for the XOR oracle applying the Deutsch construction to disjoint pairs gives the matching algorithm, so

$$Q_E(\mathrm{Par}_N)=\lceil N/2\rceil,\qquad D(\mathrm{Par}_N)=N\tag{14}$$

for ~~point-value access~~the XOR oracle [20, 21]. ~~A binary partition can therefore remain query intensive. If instead one declares~~ For the qutrit shift X_3, Theorem 1 forbids the pair construction, and $\lceil N/2\rceil\le Q_E(\mathrm{Par}_N;X_3)\le N$. Whether the upper bound is attained is open; if it is, two oracles with identical classical values would differ by a factor of two for all N.

If instead an oracle $\widetilde{O}_b|y\rangle=|y\oplus\mathrm{Par}(b)\rangle$ ~~to be a single oracle~~ that returns the parity counts as one call, parity ~~becomes one-query resolvable classically as well as~~ takes one query, classically and quantum mechanically. ~~That does not defeat~~ This does not contradict Eq. (14); ~~it replaces point access by an oracle that already aggregates the desired~~ such an oracle already returns the answer.

~~The preceding examples point to a common picture: the answer partition specifies the required outputs, whereas the query model determines the cost of separating its cells. We test this picture exhaustively for small point-oracle problems. Let the hidden instance be~~

IV. EXACT CENSUS OF BALANCED TASKS FOR THE STANDARD ORACLE

So far the access varied and the task stayed fixed. Now the standard oracle stays fixed and the task varies. The hidden instance is $b\in\{0,1\}^N$, accessed through Eq. (8) with ~~the address range enlarged to~~ N ~~, and consider every binary answer map having~~ addresses, and we consider every two-valued answer with 2^{N-1} instances in each cell. ~~Exchanging the answer labels~~ Swapping the two answers gives the same ~~unlabeled~~ partition, so ~~their number~~ the number of such partitions is

$$M_N=\frac{1}{2}\binom{2^N}{2^{N-1}}.\tag{15}$$

~~Coordinate permutations, independent complements of~~ Permuting coordinates, complementing coordinates, and ~~answer exchange preserve the point-oracle query cost. They~~ swapping the answers preserve both D and Q_E. These symmetries reduce the $M_3=35$ partitions to ~~six~~ 6 orbits and the $M_4=6435$ partitions to 58. ~~Orbit sizes are restored when reporting the counts below~~ The counts below include the orbit sizes.

~~For each partition we computed the optimal classic deterministic decision-tree depth~~ *Methods.* D ~~and sought matching bounds on the exact quantum query complexity~~ was computed exactly by dynamic programming over subcubes. By Proposition 4 (Ref. [15]), $Q_E=1$ exactly for dictators and two-bit parities. Upper bounds on Q_E ~~. One-query feasibility reduces, for the Boolean response, to rational Gram constraints and was solved over the rationals. Multiquery feasibility was formulated with the Gram-matrix semidefinite program of Barnum, Saks, and Szegedy [31]. Constructive upper bounds were retained as~~ come from adaptive trees whose nodes read either one bit or the parity of two bits~~; the latter is one Deutsch query . Such trees are useful certificates, but they are not a general characterization of~~, at one query each; such trees prove upper bounds but do not capture all exact quantum algorithms [16]. Lower bounds $Q_E\ge 3$ come from the two-query semidefinite program of Barnum, Saks, and Szegedy [31], in its

bounded-error form, which is numerically better behaved than an exact feasibility test. In the phase form of the standard oracle, $|i\rangle \mapsto (-1)^{b_i}|i\rangle$ with a null address $i=0$ and $b_0=0$, put $D_i=\mathrm{diag}_b((-1)^{b_i})$ (so $D_0=I$). A two-query algorithm with worst-case success p yields positive semidefinite $2^N\times 2^N$ matrices A_i, B_i, Γ_0, Γ_1 with $\sum_i A_i=\mathbf{1}\mathbf{1}^{\mathsf{T}}$, where $\mathbf{1}$ is the all-ones vector, $\sum_i D_iA_iD_i=\sum_i B_i$, $\sum_i D_iB_iD_i=\Gamma_0+\Gamma_1$, and $(\Gamma_{f(b)})_{bb}\ge p$. If real symmetric Y_1,Y_2,Y_3 and a probability vector λ satisfy

$$Y_1\succeq D_iY_2D_i,\quad Y_2\succeq D_iY_3D_i\quad (i=0,\dots,N),\qquad Y_3\succeq \Lambda_j\quad (j=0,1),\tag{16}$$

with $\Lambda_j=\mathrm{diag}_b\big(\lambda_b[f(b)=j]\big)$, where $[\cdot]$ is 1 if the condition holds and 0 otherwise, then, with $\langle Y,A\rangle=\mathrm{Tr}(YA)$,

$$\begin{aligned}\mathbf{1}^{\mathsf{T}}Y_1\mathbf{1}&=\textstyle\sum_i\langle Y_1,A_i\rangle\ge\langle Y_2,\sum_i B_i\rangle\ge\sum_j\langle Y_3,\Gamma_j\rangle\\&\ge\textstyle\sum_b\lambda_b(\Gamma_{f(b)})_{bb}\ge p,\end{aligned}\tag{17}$$

so $\mathbf{1}^{\mathsf{T}}Y_1\mathbf{1}<1$ proves $Q_E\ge 3$. We solved Eq. (16) numerically with a barrier method, added 3ε, 2ε, and ε times the identity to Y_1, Y_2, and Y_3 with $\varepsilon=2^{-20}$, which raises every constraint matrix in Eq. (16) by εI, rounded every entry to a dyadic rational with denominator 2^{40}, and verified all twelve matrix inequalities and $\lambda\ge 0$, $\sum_b\lambda_b=1$ in exact integer arithmetic (positive pivots in Gaussian elimination). Each certificate is a list of integers, given in the Supplemental Material, and can be checked without the solver.

~~For $N=3$ all lower and upper certificates meet exactly. Table II shows every orbit, rather than only aggregate counts~~ *Use of AI tools.* The census code and the Lean formalization were written with an OpenAI Codex assistant, and the dual-certificate solver and the exact verifier with Anthropic's Claude (versions are given in the Acknowledgments). No reported result rests on these tools alone: the census counts were reproduced by two independently written programs and agree with Ref. [16], the Lean audits reproduce the finite census, and every certificate is checked by a verifier that uses only exact integer arithmetic and does not depend on the solver. ~~Here $\mathrm{Sel}(c,u,v)=u$ for $c=0$ and v for $c=1$.~~

~~Thus 20 of the~~

Three bits. For $N=3$ the lower bounds from the one-query class match the explicit trees for every task (Table II). These values are known: Montanaro, Jozsa, and Mitchison give optimal exact algorithms for every three-bit function [16]; we only arrange them as balanced partitions. Of the 35 balanced tasks~~admit~~, 20 have an exact advantage: three by a factor of two and seventeen by a factor of 3/2. The ~~certificates also display the conditional structure hidden by the bare partition. For example,~~ trees show how such algorithms work. For three-bit majority~~is decided by first obtaining~~, first obtain $b_1\oplus b_2$; if it is ~~zero one reads~~ 0, read b_1, ~~and if it is one one reads~~ otherwise read b_3. ~~Likewise,~~ ; this is the algorithm of Ref. [16], Sec. 6.2. Likewise, for $F=b_1\oplus b_2b_3$~~is decided by first reading~~, first read b_2, ~~followed by~~ then b_1 ~~when~~ if $b_2=0$~~and by~~, or the Deutsch parity $b_1\oplus b_3$ ~~when~~ if $b_2=1$. ~~Both real multilinear polynomials have degree three, so the polynomial method excludes one query and proves these two-query constructions optimal.~~

TABLE II. Balanced three-bit tasks up to coordinate permutations, input complements, and answer exchange. The orbit sizes sum to 35. Here Par_3 and Maj_3 are the parity and the majority of the three bits, and $\mathrm{Sel}(c,u,v)=u$ for $c=0$ and v for $c=1$. The values of Q_E are not new: Ref. [16] gives explicit optimal exact algorithms for every function on up to three bits.

Representative	Orbit	D	Q_E
~~x_1~~ b_1	3	1	1
~~$x_1\oplus x_2$~~ $b_1\oplus b_2$	3	2	1
Par_3	1	3	2
Maj_3	4	3	2
~~$x_1\oplus x_2x_3$~~ $b_1\oplus b_2b_3$	12	3	2
~~$\mathrm{Sel}(x_1,x_2,x_3)$~~ $\mathrm{Sel}(b_1,b_2,b_3)$	12	2	2

Four bits. For $N=4$ ~~, exact rational lower certificates and explicit trees settle~~ the one-query class and explicit two-query trees settle 15 orbits covering 395 partitions~~: 4, 6, 48, 248, and 89 have $(D,Q_E)=(1,1),(2,1),(2,2),(3,2)$, and $(4,2)$, respectively. Of these, 343 have an exact advantage. For~~. For each of the remaining 43 orbit representatives, covering 6040 partitions, ~~exact~~ explicit trees give $Q_E\le 3$ ~~, while two SCS (Splitting Conic Solver) runs at requested tolerances 10^{-7} and 10^{-8} report the~~ and the exact rational certificates bound the optimal two-query ~~SDP infeasible. The resulting assignments, 1536 at $(D,Q_E)=(3,3)$~~ worst-case success by values between 0.9001 and ~~4504 at $(4,3)$, are therefore *numerical candidates*, not exact lower bounds. If accepted, they give candidate totals of 0.9862, hence $Q_E=3$.~~ The classification of Table III is therefore exact: 4847 ~~speedup~~ balanced four-bit partitions have an exact quantum advantage (6 by a factor of two at $(2,1)$, 89 by a factor of two at $(4,2)$, 248 by a factor of 3/2, and 4504 by a factor of 4/3) and 1588 ~~no-speedup partitions. Exact dual certificates are required before these totals can be stated as theorems~~have none. The polynomial method cannot give these lower bounds: the 43 orbits have real degree 3 (2144 partitions) or 4 (3896 partitions), so it yields only $Q_E\ge 2$.

~~Numerical SDP studies of~~ Ref. [16] reported numerical SDP values for all Boolean functions ~~through four bits were already reported in Ref. [16]; subsequent structural~~ of up to four bits and noted that such values do not by themselves prove exactness; later

TABLE III. Balanced four-bit tasks grouped by (D, Q_E). Orbits are taken up to coordinate permutations, input complements, and answer exchange; the columns sum to 58 orbits and 6435 partitions. The last column names the source of the lower bound on Q_E: Proposition 4 excludes one query, and the rational dual certificates of Eq. (16) exclude two.

(D, Q_E)	Orbits	Partitions	Representative	Lower bound
$(1,1)$	1	4	b_1	—
$(2,1)$	1	6	$b_1 \oplus b_2$	—
$(2,2)$	1	48	$\mathrm{Sel}(b_1, b_2, b_3)$	Prop. 4
$(3,2)$	8	248	$\mathrm{Maj}(b_1, b_2, b_3)$	Prop. 4
$(4,2)$	4	89	Par_4	Prop. 4
$(3,3)$	10	1536	$\mathrm{Sel}(b_1, b_2 b_3, b_2 \vee b_4)$	certificate
$(4,3)$	33	4504	$b_1 \oplus b_2 \oplus b_3 b_4$	certificate

work on exact two-query algorithms ~~also uses that four-bit numerical classification~~builds on that classification [22]. We ~~cross-validated~~ checked our orbit representatives against the published NPN (~~Negation-Permutation-Negation~~negation–permutation–negation) identifiers of Ref. [16]. After removing dummy variables, its one- and two-query entries agree with all 15 representatives ~~certified here. For the~~ settled by trees, and for the other 43 ~~remaining representatives, its reported~~ its optimal two-query success probabilities~~lie between~~ , 0.900 ~~and~~ to 0.986, ~~consistent with our infeasibility results but still numerical. This agreement checks the enumeration and symmetry bookkeeping; it does not manufacture an exact lower bound~~agree with the optimal values 0.9000–0.9861 of our dual solutions. The certificates turn this numerical agreement into exact lower bounds; the 15 orbits settled by trees give dual value one, as they must.

Lean formalization. The Supplemental Material includes a Lean 4/mathlib formalization of the exact algebraic and finite parts. Three `native_decide` checks reproduce the 35-task and 6435-task censuses of decision trees and restricted trees, including the 395 cases settled by trees; `native_decide` also trusts Lean's compiler, through an extra axiom that `#print axioms` reports. The 43 dual certificates are checked by the separate exact verifier, not in Lean. Density operators, general quantum query algorithms, and the general parity lower bound are not formalized.

The ~~narrower contribution of the census is therefore to reorganize~~ census arranges balanced functions as ~~unlabeled~~ answer partitions, ~~quotient only by symmetries that preserve point access, attach simple exact certificates where available, and keep numerical status explicit. The observed proportions are finite-model facts, not an asymptotic speeduperiterion~~identifies only partitions related by symmetries of the oracle, and attaches a short exact certificate to every class. Its proportions describe small cases only; they are not a criterion for asymptotic speedup.

~~Bernstein–Vazirani provides a complementary access-model example.~~

V. DECOHERED ACCESS: BERNSTEIN–VAZIRANI WITH AN INACCESSIBLE WITNESS

For $a, x \in \mathbb{F}_2^n$, let $f_a(x) = a \cdot x \pmod 2$ and $O_a |x, y\rangle = |x, y \oplus a \cdot x\rangle$. With ordinary phase ~~access, the corresponding character states~~ kickback the states for different a are orthogonal, so ~~the singleton partition is resolved in one query~~ one query identifies a [32]. We now ~~add an inaccessible address record while preserving every computational-basis value returned by the oracle~~let the oracle also leave an inaccessible record of the address, without changing any value it returns on basis states.

Choose $S \subseteq \{1, \dots, n\}$ ~~,~~with $|S| = k$, ~~and~~ let x_S ~~denote~~ be the corresponding substring ~~. Consider the unitary dilation~~

$$W_a^S |x, y, e\rangle = |x, y \oplus a \cdot x, e \oplus x_S\rangle .$$

~~Each call receives a fresh witness~~ of x, and let $\bar{S}$ be the complement of S. Each call acts as

$$W_a^S |x, y, e\rangle = |x, y \oplus a \cdot x, e \oplus x_S\rangle \tag{18}$$

on the address x, the target bit y, and a fresh k-bit witness register E with content e, prepared in $|0^k\rangle$ ~~, which is discardedafterward. The accessible oracle is consequently the channel~~

$$\Phi_a^S(\rho) = \mathrm{Tr}_E\left[W_a^S(\rho \otimes |0^k\rangle\langle 0^k|)(W_a^S)^\dagger\right] .$$

and then discarded. The oracle the algorithm sees is therefore the channel

$$\Phi_a^S(\rho) = \mathrm{Tr}_E\left[W_a^S(\rho \otimes |0^k\rangle\langle 0^k|)(W_a^S)^\dagger\right] = O_a\, \Delta_S(\rho)\, O_a^\dagger, \tag{19}$$

where $\Delta_S(\rho) = \sum_s \Pi_s \rho \Pi_s$ dephases the address register in the coordinates S (Π_s projects onto $x_S = s$); copying x_S into a discarded register is exactly this dephasing, and O_a commutes with every Π_s. Basis queries still return $a \cdot x$ exactly, ~~while the witness records selected address~~

information and removes coherences between sectors but the witness destroys coherence between addresses with different x_S, as in standard which-way interference [23] . Define the surviving answer partition experiments [23] and in the coherence analysis of the Bernstein–Vazirani algorithm [24].

Surviving partition. Define

$$\Pi_S = \{C_u : u \in \mathbb{F}_2^{n-k}\}, \qquad C_u = \{a : a_{\bar{S}} = u\}. \tag{20}$$

The one-query statement below concerns this surviving partition Π_S, whose cells are indexed by $a_{\bar{S}}$; it is therefore an access-induced change of the task-oracle pair rather than a same-partition comparison for the original singleton task. For the classical comparison, let $\mathcal{O}^{\mathrm{val}} = \{O_a^{\mathrm{val}} : a \in \mathbb{F}_2^n\}$ denote the value-oracle family induced by computational-basis preparation and readout of be the value oracle obtained by preparing and reading out Φ_a^S in the computational basis: a query chooses x and returns $a \cdot x$. For $k < n$ and worst-case error $\epsilon < 1/2$, these explicitly specified quantum and classical interfaces give

$$Q_\epsilon^\Phi(\Pi_S) = 1, \qquad R_\epsilon(\Pi_S; \mathcal{O}^{\mathrm{val}}) = n - k. \tag{21}$$

For the quantum part, the usual Bernstein–Vazirani input and phase kickback, followed by tracing out the witness, leave Eq. (19), leave on the n-qubit address register (after factoring out the target qubit, which stays in $|-\rangle$) the state

$$\rho_a^S = 2^{-n} \sum_{x,x' :\, x_S = x'_S} (-1)^{a\cdot(x\oplus x')} \, |x\rangle\langle x'| \, . \tag{22}$$

Character orthogonality after the final Hadamards gives, up to a fixed reordering of coordinatesAfter the final Hadamard gates, the orthogonality of characters gives, with the coordinates reordered and z the measured address string,

$$\begin{aligned} H^{\otimes n} \rho_a^S H^{\otimes n} &= |a_{\bar{S}}\rangle\langle a_{\bar{S}}| \otimes \frac{I_S}{2^k}, \\ \Pr(z|a) &= 2^{-k}\, \mathbf{1}[z_{\bar{S}} = a_{\bar{S}}]. \end{aligned} \tag{23}$$

Thus the one-query readout exactly So one query resolves Π_S . Zero queries cannotresolve this nontrivial partition Π_S, because exactly; zero queries cannot, since the output would be independent of its celllabel. The classical upper bound uses the not depend on the cell. Classically, the $n-k$ basis queries outside S suffice. For the classical lower bound, impose the favorable grant even the promise $a_S = 0$. Under the uniform distribution on and let $a_{\bar{S}}$, fewer be uniformly distributed. Fewer than $n-k$ returned linear equations leave an affine solution space of positive dimension, so even a decoder given the complete at least two equally likely candidates, so a decoder that sees the whole transcript succeeds with probability at most 1/2. The same average bound holds after including Averaging over the protocol's internal randomness, and therefore random choices does not change this, so some input has success at most 1/2. If the witness were coherently available

Identifying a. The same channel also changes the cost of the original task, identifying a, that is, resolving the partition Π_{single} into single instances.

Proposition 6 (Witness cost of identification)**.** *For $0 \le k \le n$, with S known to the algorithm, the exact quantum cost of identifying a through $\{\Phi_a^S\}$ is $Q_E^\Phi(\Pi_{\mathrm{single}}) = \min\{k+1, n\}$; classically, $R_\epsilon(\Pi_{\mathrm{single}}; \mathcal{O}^{\mathrm{val}}) = n$ for $\epsilon < 1/2$.*

Proof. Upper bound: one Bernstein–Vazirani query with $x_S = 0$ returns $a_{\bar{S}}$, and k basis queries $x = e_i$, $i \in S$, return a_S; alternatively, n basis queries return a. At $k = 0$ this is the ordinary one-query algorithm. Lower bound for $k \ge 1$: record each dephasing in Eq. (19) in an inaccessible environment register. For a record $\vec{s} = (s_1, \ldots, s_T)$, $s_t \in \mathbb{F}_2^k$, let $|\phi_a(\vec{s})\rangle$ be the conditional unnormalized output, so that $\rho_a = \sum_{\vec{s}} |\phi_a(\vec{s})\rangle\langle\phi_a(\vec{s})|$. Because the records are inaccessible, exact identification requires $\langle\phi_a(\vec{s})|\phi_{a'}(\vec{s}')\rangle = 0$ for all $a \ne a'$ and all records $\vec{s}, \vec{s}'$. In the target qubit's X basis, the tth query multiplies the component at address $x = (s_t, z_t)$, where $z_t = x_{\bar{S}}$ holds the unrecorded coordinates, either by 1 or by $(-1)^{a\cdot(s_t,z_t)}$. Hence $\phi_a(\vec{s})$ is a linear combination of characters $(-1)^{a\cdot f}$ whose frequencies are $f = 0$ or $f = \sum_{t\in F}(s_t, z_t)$ for nonempty subsets F of the queries. (i) If $s_1, \ldots, s_T$ do not span $\mathbb{F}_2^k$, choose $h \ne 0$ orthogonal to all s_t. Every such character is then invariant under $a \mapsto a + (h, 0)$, so $\phi_{a+(h,0)}(\vec{s}) = \phi_a(\vec{s})$, and orthogonality forces $\phi_a(\vec{s}) = 0$. Hence $T \ge k$, and only records whose s_t span $\mathbb{F}_2^k$ contribute. (ii) Let $T = k < n$. In every contributing record the s_t are linearly independent, so every frequency with $F \ne \emptyset$ has a nonzero S part, and for $u \ne 0$ the sum $\sum_a (-1)^{u\cdot a_{\bar{S}}} \phi_a(\vec{s})$ vanishes. Since the $\phi_a(\vec{s})$ are pairwise orthogonal, the squared norm of this sum is $\sum_a \|\phi_a(\vec{s})\|^2$, so $\phi_a(\vec{s}) = 0$ for every contributing record, contradicting normalization. Hence $T \ge k+1$ for $k < n$. The classical bound is the singleton case of the argument above. □

At $k = 0$ nothing is recorded, and Eq. (23) is the ordinary Bernstein–Vazirani output, with one quantum query against n classical ones. Each recorded address bit adds exactly one query until the cost reaches the classical value n; from $k = n-1$ on, the quantum advantage for identification is gone, while the classical cost stays n. If the witness were available to the algorithm, the fixed coupling $|x, e\rangle \mapsto |x, e \oplus x_S\rangle$ could be undone without another oracle call, restoring the ordinary Bernstein–Vazirani output. The accessible oracle output. What the algorithm accesses here is a channelrather than the globally unitary dilation, highlighting the distinction between standard and garbage-bearing oracle access in quantum query complexity [33], not the unitary W_a^S: as with in-place versus standard oracles [4] and faulty

oracles [7], whether the oracle leaves an inaccessible record is part of the problem.

~~At $k = 0$, no address coordinate is recorded and~~

VI. DISCUSSION

All results of this paper follow one strategy, which can be stated in four steps; the physics enters in the middle two.

- *Answer partition* Π_a [Eq. ~~(23) is the ordinary Bernstein–Vazirani output resolving the singleton partition, giving the n-versus-one identification gap. At $k = n$, the stated interference readout is uniform and Π_S has only one cell, although ordinary basis queries can still recover a. The model separates loss of the phase-interference route from loss of the classical function values themselves~~(1)]: Put two hidden instances in the same cell whenever the task has the same correct answer for both. The cells C_j are sets of classical instances, not subspaces. The algorithm must learn the cell, and only the cell: in Deutsch's problem the four possible functions fall into two cells, constant and balanced, and the algorithm never has to find out which function within a cell it faces.

- *Oracle access* (Secs. III and V): Fix what one oracle call does. The partition fixes the correct answers to the question; the access model fixes how that answer can be obtained by quantum state discrimination.

  ~~We conclude by returning to the role of the answer partition. Equation (1) specifies which hidden instances share an answer, while the oracle and protocol provide the physical ingredients—the probe state, oracle transformation, accessible output space, and measurement—that resolve the corresponding answer classes.Together, these ingredients determine how rapidly states from different cells separate. The response-spectrum theorem and the contrast between point and aggregate parity show how changing the access model can change this separation even when the abstract task remains fixed. Conversely,~~

- *Probe and one quantum oracle call* (Sec. II): Choose an instance-independent probe state, for instance a uniform superposition obtained by "fanning out" a single basis state with Hadamard gates. One oracle call turns the probe into the output state $|\Psi_a\rangle$ of the hidden instance a.

- *Orthogonality across cells* (Proposition 1): Ask whether some probe makes $\langle\Psi_a|\Psi_{a'}\rangle = 0$ whenever $g(a) \neq g(a')$ [Eq. (5)]. A negative answer needs no search over the infinitely many probes: a single matrix, a dual certificate (Proposition 2), proves that none works. Only orthogonal states can be told apart with certainty, so exactness requires orthogonality, but only between cells. Within a cell, states may overlap or even coincide: the measurement reads the cell label, not the instance, as a post office sorts letters by postal code without reading the street.

Two equally weighted branches of a quantum state cancel, that is, interfere destructively, only if one carries an extra factor -1 relative to the other. In the examples of the Introduction, the ~~finite census holds the point oracle fixed while varying the balanced target. Its exact three-bit certificates show how quantum advantages can arise from oracle-accessible, conditionally composed refinements, while the four-bit calculation clearly separates exact certificates from numerical candidates.~~ standard flip on a qubit and the transposition on a qutrit both have the eigenvalue -1 and separate constant from balanced functions in one call; the qutrit cyclic shift, whose eigenvalues are the cube roots of unity, cannot (Theorem 1 and Corollary 1). With more than two branches, other phases can cancel, for instance $1 + \omega + \omega^2 = 0$ for $\omega = e^{2\pi i/3}$; this is the interference used by the quantum Fourier transform and Shor's algorithm. Deutsch's problem, however, compares only two address branches, and therefore needs -1. With more calls, the same orthogonality is required of the final states, with instance-independent operations between the calls (Sec. IV); with bounded error, it is relaxed to a required success probability (Proposition 5).

Some of these separations need no non-Clifford operations. The standard Deutsch and Bernstein–Vazirani protocols, including their XOR oracles, use only Clifford operations, and so does the witness coupling of Eq. (18) before the witness is discarded. Given a full classical description of the circuit, they can therefore be simulated efficiently [34, 35]. In the query model, however, such a description would reveal the hidden oracle data, which is more than calling the oracle provides. For these examples, nonstabilizer resources ("magic" [36]) are not needed for a query advantage. General oracles of this kind with a binary-encoded address need multiply controlled gates, which are not Clifford, so this remark is specific to Deutsch and Bernstein–Vazirani.

The results vary these steps one at a time. Corollary 1 and Proposition 6 change the access (the second step) while holding the task and its classical values fixed, and the exact cost changes; Proposition 5 shows what survives with bounded error. Theorem 1 and Proposition 4 give the rule for every response unitary. The sign oracle of Proposition 3 is a deliberately weaker oracle, whose values are invisible to basis queries. The census keeps the standard oracle fixed and varies the balanced task (the first step), and the cost changes again: for $N \leq 4$ every class is certified exactly. Shor's and Grover's algorithms [37, 38] fit the same ~~task–oracle~~

perspective: Shor exploits periodic structure in picture: each exploits structure of the oracle family accessible via the quantum Fourier transform, while Grover exploits symmetry and coherent amplitude amplification via a matched phase oracle. These examples reinforce that quantum speedup depends on the structure of the oracle access, rather than on the coarseness or cardinality of the answer partition alonethat matches a specific coherent access.

Partitions are therefore indispensable as specifications but insufficient as complexity criteria. Quantum advantage is not caused by partition coarseness or by simultaneous access to imagined branch values. It occurs when structure-matched interference separates the required needed to state a task but do not decide its complexity, and neither do classical values. A quantum advantage occurs when interference matched to the oracle's structure separates the answer cells in fewer calls than any classical transcript under the same interfaceprotocol needs with the same access. Superlinear exact separations for total Boolean functions are known [39], but establishing any an asymptotic speedup requires a structured family of task–oracle pairs; percentages from a finite census cannot substitute for replace that analysis. Within the present framework, the parity question for the qutrit shift (Sec. III) is a natural test of whether a response-spectrum effect persists for all N. A second open question is which response V is optimal for several queries. A response with eigenvalues 1 and -1 reproduces the XOR oracle exactly in each call, by encoding the control qubit in the corresponding eigenvectors, and the degree bound holds for every V; whether some other response beats the XOR oracle for some task with two or more queries remains open.

In summary, the paper adds three things. First, a complete one-query classification for every response unitary (Theorem 1 and Proposition 4), which extends the XOR-oracle characterization of Chen, Ye, and Li [15]. Second, a caution for implementations: "a quantum oracle for f" is underspecified, since reversible oracles with the same classical readout can differ in quantum cost. This matters in particular for qudit realizations, where the natural reversible extension may be a cyclic shift without the eigenvalue -1; the one-query solution of Deutsch's problem is then lost. Third, exact results where only numerical ones existed: rational certificates for the 6040 four-bit cases, the optimal bounded-error success of odd cyclic responses, and the exact cost $\min\{k+1, n\}$ of identification with an inaccessible address copy. All effects found here are constant factors; none is an asymptotic speedup.

ACKNOWLEDGMENTS

The author thanks Noson S. Yanofsky for discussions that motivated this work. The author usedan OpenAI Codex language-model-based assistant for Two AI assistants were used. OpenAI Codex ChatGPT Astra ultra supported structural editing, consistency checks, and the implementation of the finite partition census , census code, and the Lean 4 formalization. Claude Opus 5.5 (Anthropic) was used in the revision to review the manuscript and the literature, to draft revised text—including the statements and proofs of Propositions 2–6 and the new proof of Theorem 1—and to carry out the bounded-error and accepts dual-certificate computations of Secs. III and IV. The author directed both tools, checked the mathematical arguments, and verified the computations as described in Sec. IV; the author accepts full responsibility for the final scientific content content and code. This research was funded in whole or in part by the Austrian Science Fund (FWF), grant PIN5424624. The author acknowledges TU Wien Bibliothek for support through its Open Access Funding Programme.

Data and code availability. The source code, exact certificates See Supplemental Material at [URL will be inserted by publisher] for the Python and Lean 4 sources, the proof report, the 43 rational dual certificates with an exact verifier, the census tables and witness files, numerical witnesses, generated tables , and the NPN cross-validation underlying the finite census are available at https://svozil.github.io/publications/2026-qc-partition-census-suppl the published NPN data of Ref. [16].

[1] D. Deutsch, Quantum theory, the Church-Turing principle and the universal quantum computer, Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences **400**, 97 (1985).

[2] D. Deutsch and R. Jozsa, Rapid solution of problems by quantum computation, Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences **439**, 553 (1992).

[3] R. Cleve, A. Ekert, C. Macchiavello, and M. Mosca, Rapid solution of problems by quantum computation, Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences **454**, 339 (1998).

[4] E. Kashefi, A. Kent, V. Vedral, and K. Banaszek, Comparison of quantum oracles, Physical Review A **65**, 050304 (2002), quant-ph/0109104.

[5] A. Wójcik and R. W. Chhajlany, Quantum-classical correspondence in the oracle model of computation, International Journal of Quantum Information **4**, 633 (2006), arXiv:quant-ph/0506248.

[6] C. S. Calude, De-quantizing the solution of Deutsch's problem, International Journal of Quantum Information **5**, 409 (2007), quant-ph/0610220.

[7] O. Regev and L. Schiff, Impossibility of a quantum speed-up with a faulty oracle, in *Automata, Languages and Programming, ICALP 2008*, Lecture Notes in Computer Science, Vol. 5125 (Springer, 2008) pp. 773–781, 1202.1027.

[8] N. Donath and K. Svozil, Finding a state among a complete set of orthogonal ones, Physical Review A **65**, 044302 (2002), arXiv:quant-ph/0105046.

[9] K. Svozil, Quantum information in base n defined by state partitions, Physical Review A **66**, 044306 (2002), arXiv:quant-ph/0205031.

[10] K. Svozil, Quantum information via state partitions and the context translation principle, Journal of Modern Optics **51**, 811 (2004), arXiv:quant-ph/0308110.

[11] K. Svozil, Characterization of quantum computable decision problems by state discrimination, in *Quantum theory: reconsideration of foundations—3*, AIP Conf. Proc., Vol. 810 (Amer. Inst. Phys., Melville, NY, 2006) pp. 271–279.

[12] K. Svozil and J. Tkadlec, On the solution of trivalent decision problems by quantum state identification, Natural Computing **8**, 539 (2009).

[13] K. Svozil, Orthogonal vector computations, Entropy **18**, 156 (2016), arXiv:1601.07106.

[14] A. M. Turing, Systems of logic based on ordinals, Proceedings of the London Mathematical Society, Series 2 **45**, 161 (1939).

[15] W. Chen, Z. Ye, and L. Li, Characterization of exact one-query quantum algorithms, Physical Review A **101**, 022325 (2020), 1912.03493.

[16] A. Montanaro, R. Jozsa, and G. Mitchison, On exact quantum query complexity, Algorithmica **71**, 775 (2015), arXiv:1111.0475.

[17] A. Acín, Statistical distinguishability between unitary operations, Physical Review Letters **87**, 177901 (2001), arXiv:quant-ph/0102064.

[18] R. Duan, Y. Feng, and M. Ying, Perfect distinguishability of quantum operations, Physical Review Letters **103**, 210501 (2009), arXiv:0908.0119.

[19] Z. Ye and L. Li, Characterization of exact one-query quantum algorithms for partial Boolean functions, Journal of Computer Science and Technology **38**, 1423 (2023), 2008.11998.

[20] R. Beals, H. Buhrman, R. Cleve, M. Mosca, and R. de Wolf, Quantum lower bounds by polynomials, Journal of the ACM **48**, 778 (2001), arXiv:quant-ph/9802049.

[21] E. Farhi, J. Goldstone, S. Gutmann, and M. Sipser, Limit on the speed of quantum computation in determining parity, Physical Review Letters **81**, 5442 (1998), arXiv:quant-ph/9802045.

[22] S. Ye, W. Yang, and L. Huang, Characterization of exact two-query quantum algorithms, Information and Computation **298**, 105166 (2024).

[23] B.-G. Englert, Fringe visibility and which-way information: An inequality, Physical Review Letters **77**, 2154 (1996).

[24] M. Naseri, T. V. Kondra, S. Goswami, M. Fellous-Asiani, and A. Streltsov, Entanglement and coherence in the Bernstein–Vazirani algorithm, Physical Review A **106**, 062429 (2022).

[25] A. Ambainis, Quantum lower bounds by quantum arguments, Journal of Computer and System Sciences **64**, 750 (2002).

[26] T. Lee, R. Mittal, B. W. Reichardt, R. Špalek, and M. Szegedy, Quantum query complexity of state conversion, in *2011 IEEE 52nd Annual Symposium on Foundations of Computer Science* (IEEE, 2011) pp. 344–353.

[27] B. W. Reichardt, Span programs are equivalent to quantum query algorithms, SIAM Journal on Computing **43**, 1206 (2014).

[28] M. Araújo, A. Feix, F. Costa, and Č. Brukner, Quantum circuits cannot control unknown operations, New Journal of Physics **16**, 093026 (2014), arXiv:1309.7976.

[29] J. L. Cereceda, Generalization of the Deutsch algorithm using two qudits (2004), arXiv:quant-ph/0407253.

[30] D. A. Meyer and J. Pommersheim, On the uselessness of quantum queries, Theoretical Computer Science **412**, 7068 (2011), arXiv:1004.1434.

[31] H. Barnum, M. Saks, and M. Szegedy, Quantum query complexity and semi-definite programming, in *Proceedings of the 18th IEEE Conference on Computational Complexity* (IEEE Computer Society, 2003) pp. 179–193.

[32] E. Bernstein and U. Vazirani, Quantum complexity theory, SIAM Journal on Computing **26**, 1411 (1997), preliminary version in *Proceedings of the 25th Annual ACM Symposium on Theory of Computing* (STOC 1993), pp. 11–20.

[33] S. Aaronson, Open problems related to quantum query complexity, ACM Transactions on Quantum Computing **2**, 15 (2021), arXiv:2109.06917.

[34] D. Gottesman, The Heisenberg representation of quantum computers, in *Group22: Proceedings of the XXII International Colloquium on Group Theoretical Methods in Physics*, edited by S. P. Corney, R. Delbourgo, and P. D. Jarvis (International Press, Cambridge, MA, 1999) pp. 32–43, arXiv:quant-ph/9807006.

[35] S. Aaronson and D. Gottesman, Improved simulation of stabilizer circuits, Physical Review A **70**, 052328 (2004).

[36] V. Veitch, S. A. H. Mousavian, D. Gottesman, and J. Emerson, The resource theory of stabilizer quantum computation, New Journal of Physics **16**, 013009 (2014).

[37] P. W. Shor, Algorithms for quantum computation: discrete logarithms and factoring, in *Proceedings of the 35th Annual Symposium on Foundations of Computer Science (FOCS '94)* (IEEE Computer Society Press, Los Alamitos, CA, USA, 1994) pp. 124–134, arXiv:quant-ph/9508027.

[38] L. K. Grover, A fast quantum mechanical algorithm for database search, in *Proceedings of the Twenty-Eighth Annual ACM Symposium on the Theory of Computing* (1996) pp. 212–219, arXiv:quant-ph/9605043.

[39] A. Ambainis, Superlinear advantage for exact quantum algorithms, SIAM Journal on Computing **45**, 617 (2016), arXiv:1211.0721.